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Phonetics, phonemics, and morphophonemics

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PIE mobile accent in Italic: Further evidence

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Several recent proposals have challenged the handbook assumption that the Proto-Indo-European mobile accent was lost in Italic, such as Rix's suggestion (1996, updating a theory of Hirt) that Latin apocope was sensitive to the PIE accent (cf. *et* < **éti* vs. *pede* < **ped-í*), or that Thurneysen-Havet's Law (**-ou̯(H)V-* > Ital. /-au̯V-/) operated on unaccented **o* in terms of the PIE accent (e.g. U. *sauitu* < **ksou̯-éi̯e/o-,* Lat. *cavus* < **ĥou̯H-ó-*; Vine 2004, 2006). Here I suggest that additional evidence of this kind may be seen in the problematic *ī*-conjugation of two Italic morphological categories, namely the acrostatic iterative-causative (seen in Lat. *sōpīre*) and *o*-stem denominatives (of the type *servus* → *servīre*). Both can be explained by a prehistoric raising of **e* to /i/ before **i̯* (similar to the "*pius*-Gesetz"), which occurred in unaccented position in terms of the PIE accent (**su̯ópei̯e/o-* > /su̯ōpiie-/, **seru̯-e-i̯é/ó-* > /seru̯iie-/). Accented **e* in this position is not affected (as in standard iterative-causative **-éi̯e/o-*, the "material adjective" suffix **-éi̯o-*, and originally proterokinetic or hysterokinetic *i*-stem nom. pl. **-éi̯-es*), whereas acrostatic *i*-stem nom. pl. **-ei̯-es* (with root accent) would undergo the rule (> /-iyes/), providing an attractive explanation for the Italic *i*-stem nom. pl. *-īs*.

1 Italic accent: background

1.1 According to standard handbook-style assumptions, the Proto-Indo-European mobile accent is said to have been lost without a trace in Italic, having been converted (whether directly or indirectly) to an initial-stress accent, as seen in the so-called Common Italic "period of initial

stress".[1] This, in turn, is supported with reference to well-documented phonological features like widespread syncope in Italic, Middle and Late Etruscan syncope, vowel weakening in Latin (and to a lesser extent in Sabellic), as well as certain orthographic features (such as the "geminatio vocalium" so prominent in Oscan and early Latin, largely restricted to initial syllables).[2] The Classical Latin penultimate stress system thus has no connection with pre-Italic accent, and is all but universally regarded as a still more recent development.

1.2 In recent years, however, this picture has been changing. Some handbooks now offer a more nuanced perspective that allows for possible traces of an early period of mobile accent in Italic. The main phenomena in question can be listed briefly as follows:

1 According to a widespread view,[3] there is a double reflex of syllabic resonant followed by laryngeal when these are interconsonantal, depending on whether or not the resonant was accented: unaccented CR̥HC > Ital. /CRāC/ (e.g. **pl̥h₂-nó-* > Lat. *plānus* 'flat'), but accented CŔ̥HC (typically arising from a substantivizing accent retraction) > Ital. /CaRaC/ (e.g. **pĺ̥h₂-meh₂* > **palamā* > Lat. *palma* 'palm (of the hand)', i.e. 'flat part (of the hand)', to the same root as *plānus*).[4]
2 According to another proposal,[5] final unaccented short vowels were apocopated (e.g. **éti* [cf. Ved. *áti,* Gk. *ἔτι*] > Lat. *et* 'and'), whereas accented final vowels remain (e.g. loc. sg. **ped-í* [cf. Gk. dat. *ποδί*] > Lat. abl. *pede* 'foot').

1 See e.g. Sommer-Pfister 1977: 73, Sihler 1995: 239, Meiser 1998: 53, Baldi 2002: 269, Weiss 2009: 109, Fortson 2010: 278, 284; additional references in Nishimura's recent discussion of the prehistoric initial stress (2008: 6ff.).
2 General discussion and copious earlier references: Nishimura 2008, Ch. 1.
3 See Meiser 1998: 108f., Weiss 2009: 110.
4 An exact analogue of this process is often assumed for Greek (Rix 1992: 73, Weiss 2009: 52), as in **pĺ̥h₂-meh₂* > *παλάμη* 'palm'.
5 Rix 1996: 158 n. 7 (an updated version of a theory of Hirt, see Nishimura 2008: 6 n. 14), cf. Meiser 1998: 74, Weiss 2009: 147 n. 77.

3 I have proposed another such hypothesis (2006a), according to which "Thurneysen-Havet's Law"[6] converts prevocalic **ou̯* to Ital. /au̯/ in unaccented position (in terms of PIE accentual patterns), as in iterative-causatives (e.g. **g^{wh}ou̯-éi̯e/o-* > Lat. *favēre* 'be favorable'), or "*τομός* forms" (e.g. **k̂ou̯H-ó-* 'swelling' > Lat. *cavus* 'hollow'), but not otherwise (cf. acrostatic **h$_2$óu̯-i-* 'sheep' > Lat. *ovis*).[7]

4 Note finally the conception of Latin medial vowel weakening by Pultrová (2006), who claims that medial vocalism reflects phonological treatments of PIE unaccented zero grades in certain consonantal configurations, and analogical extensions thereof.[8]

1.3 The preceding proposals may or may not be correct, and there is no space (and no need) to argue these matters here.[9] They testify, however, to a growing openness toward a conception of Italic as a branch of Proto-Indo-European that – like Germanic – may have retained indirect traces of the old PIE mobile accent. The remainder of this paper offers an additional proposal that may point in the same direction, based on two problematic morphological categories in Italic, both displaying anomalous or unexpected *ī*-conjugation: acrostatic iterative-causatives, as seen in Lat. *sōpīre* 'put to sleep' (**2**); and *o*-stem denominative verbs (**3**).

6 Schrijver 1991: 436ff. *in extenso*; earlier literature in Collinge 1985: 193ff.

7 Thurneysen-Havet's Law would in this account need to be extremely early, which seems to be the case: cf. Vine 2004: 622ff. on its operation in Sabellic (U. *sauitu* 'she shall wound' < **ksou̯-éi̯e-tōd,* to **kseu̯-* 'scrape, scratch'), and more generally de Vaan's assignment of the process to his "category B", i.e. changes occurring "[b]efore Proto-Italic split up" (2008: 8).

8 I leave aside here "Dybo's Law", which (if it exists) may reflect a broader Italo-Celto-Germanic phenomenon. See in general Meiser 1998: 75, and especially the recent discussion of Weiss 2009: 99, with further references (to which can be added Ringe 2006: 79, with a negative assessment).

9 In some ways the most radical theory is that of Pultrová (2006), **1.2**(4): see the critique by Hartmann (2008) and the comments by Weiss (2009: 121 n. 22).

2 Acrostatic iterative-causatives and Lat. *sōpīre*

2.1 The standard theory of acrostatic iterative-causatives – i.e., non-canonical iterative-causatives that correlate with acrostatic (or "Narten") root athematic presents – is based on Klingenschmitt's celebrated study of Lat. *sōpīre* 'put to sleep' (1978). According to this theory, which is promulgated as doctrine in LIV and otherwise very widely accepted,[10] Lat. *sōpīre* – with its isolated lengthened *ō*-grade root vocalism – is a deverbative relic form, reconstructible as **su̯ṓp-(i)i̯e/o-*. The accented long *ō*-grade of the root, together with the plain **-i̯e/o-* suffix (as opposed to canonical iterative-causatives with the shape R(o)-*éi̯e/o-*), suggest a special type of iterative-causative in R(ṓ)-*i̯e/o-*, comparable to and morphologically associated with acrostatic presents, which show R(ḗ) in their active strong forms.[11]

2.2 The two pillars of Klingenschmitt's analysis were, first of all, an attractive comparison with ON *sœfa* 'kill' (originally *'put to sleep') < Gmc. **su̯ōfi̯a-*; and second, possible evidence (mainly from Indo-Iranian) that the root **su̯ep-* 'sleep' formed a PIE acrostatic present **su̯ḗp-ti/*su̯ép-n̥ti*. Initially, skepticism was voiced on both of these points: thus Bammesberger (1980) offered an inner-Germanic analogical account of the vocalism of ON *sœfa*; and Jamison (1982/3), reviewing the Indo-Iranian material, showed that the unexpected full grades there, instead of pointing to original acrostatic inflection, could be secondary, and she suggested that the root **su̯ep-* itself might not even have made a present formation of any kind in the parent language. Current opinion remains somewhat unsettled. Thus Bammesberger's conception is followed by García García (2005: 18 n. 8), as against Malzahn (2007: 237 n. 1), who asserts that "*sœfa* must undoubtedly be an archaic formation, i.e., derived from a Proto-Gmc. preform with root vowel **-ō-*". On the second point, concerning **su̯ep-*: LIV

10 Recent treatments include Kümmel 2004: 139, Martzloff 2006: 677ff., Malzahn 2007: 237 n. 1.

11 Several attempts to explain the "deep" morphological background of Lat. *sōpīre* and similar forms at the level of early PIE or pre-PIE (e.g. Rasmussen 1989: 145ff., in terms of an original infixed formation; or Pooth 2009: 241, 244, 246, in terms of an "Early PIE proto-middle" **su̯ṓp(i)*) nevertheless assume, as a descriptive fact, "acrostatic" iterative-causatives of essentially the traditional sort at the level of PIE.

(s.v. **su̯ep-*, n. 6) considers the acrostatic present to be a possibility, but admits (citing Jamison) that it cannot be confirmed by the Vedic material. Note also that, while the equation between Lat. *sōpīre* and ON *sœfa* remains attractive, the "Klingenschmitt iterative-causative" is very rarely attested: LIV tallies twenty-four possible instances,[12] of which only half are deemed "sicher" – and only half of these have corresponding acrostatic presents.[13]

V. Martzloff (2006: 677) has usefully raised several additional points concerning ON *sœfa*. He observes, first, that the voiced *-f-* of this form has little significance, as Gmc. **-f-* in Old Norse can be secondarily voiced intervocalically. Further, as is well known, PIE **-éi̯e/o-* is indistinguishable in Germanic from plain **-(i)i̯e/o-*. Thus ON *sœfa* cannot unambiguously show whether Verner's Law has applied, or whether the suffix is actually plain **-(i)i̯e/o-*; therefore, the form can in principle reflect any of the preforms **su̯ŏp-ii̯e/o-, *su̯ŏp-ei̯e/o-, *su̯ōp-ii̯é/ó-, *su̯ōp-éi̯e/o-*.

2.3 I suggest further that the Klingenschmitt/LIV doctrine of acrostatic causatives is vulnerable to still other criticisms, beyond the initial reservations expressed by Bammesberger and Jamison, and in addition to the indeterminacies noted by Martzloff.[14]

On purely theoretical grounds, Jasanoff (2009: 48f.), commenting on ablauting suffixes in the verbal system, points out that the athematic optative suffix **-i̯eh₁-/*-ih₁-* "*was the only suffixed tense-aspect sign or mood sign in the entire PIE verbal system to show ablaut*" (emphasis Jasanoff), and states that "the lesson to be learned from the special status of the optative is cautionary": there is, as he notes, a whole series of problematic categories for which scholars have posited a single verbal morpheme with apophonic variants, such as LIV's "essive" in **-h₁-i̯e/o-* vs. "fientive" in **-eh₁-*,[15] or a series of Leiden School *i*-present theories.[16] Indeed, another such in-

12 See also the listing by Rasmussen (1989: 146ff.).

13 This is to be compared with a total of about thirty acrostatic presents that LIV considers reconstructible for PIE.

14 Cf. de Vaan 2008: 575 (s.v. *sopor*): "the existence of this verbal type is uncertain".

15 Following Harðarson 1998 (but see Jasanoff 2002/3, 2003: 155ff.; Ruijgh 2004).

16 Jasanoff cites Kortlandt 1990: 7 for **-ei- ~ *-i-* (Balto-Slavic); cf. further Kloekhorst 2006 and 2008 (*passim*) for pre-Hittite **-oi- ~ *-i-*, and Schrijver 2003

stance, as Jasanoff notes, is Klingenschmitt's zero-grade causative suffix *-i̯e/o- beside the ordinary full grade *-éi̯e/o-. In Jasanoff's words: "Even if there were no straightforward alternatives to these proposals, the near-absence of ablauting suffixes elsewhere in the verbal system would justify a position of skepticism" (2009: 49).

More concretely, underlying problems inherent in the theory emerge from consideration of the LIV presentation of the material. Prime among these is a conspicuous prevalence of suffix "substitutions" or "normalizations", of which the following examples (drawn from LIV's tally of "certain" cases) are typical:

1 *g(ʷ)eĝʰ- 'eindringen, hineinwaten', caus. *gṓĝʰ-i̯e-: [SCr. *gȁziti* 'treten, waten', "Mit Suffixersatz *-i̯e- → *-ei̯e- (> -i-)";
2 *kʷeǩ- 'sehen, erblicken', caus. *kʷṓǩ-i̯e-: [AV *sáṃ kāśáyati* 'macht sichtbar', "Suffix nach dem produktiven Kausativtyp normalisiert";[17]
3 *sed- 'sich setzen', caus.-it. *sṓd-i̯e-: [OIr. *sádid* 'steckt', "Wohl aus *sōd-iei̯e- zu *sādī- umgebildet"; and [OCS *saditi* 'setzen', "Mit sekundärer *ei̯e-Flexion, falls nicht erst innerslav. Bildung";
4 *u̯edʰh₁- 'stoßen', iterat. *u̯ṓdʰh₁-i̯e-: *ὠθέω* 'stoße, dränge', and ?[OCS *vaditi* 'anklagen etc.', "Zu *u̯ōdʰ(h₁)-éi̯e- umgebildet".

Gk. *ὠθέω* in (4) is particularly instructive. The root may in fact be aniṭ;[18] and in that case, the LIV reconstruction with *-h₁- and zero-grade suffix is essentially a "trick": there is little reason (other than a wish to squeeze the form into the R(ṓ)-i̯e/o- mold) to interpret *ὠθέω* as anything other than what it appears to be – i.e., an iterative with the ordinary *-éi̯e/o- suffix (though, to be sure, with lengthened *ō*-grade in the root). Another case of this kind:

5 Lat. *cōnārī* 'attempt' < *kōnai̯e-, the result of an alleged "iterat. *kṓnh₁-i̯e-" (s.v. *kenh₁- 'sich bemühen, sich anstrengen'); but there is no other evidence that the root was seṭ (or that the laryngeal was *h₁, as LIV admits, n. 1). Further, the root not only lacks an acrostat-

(followed by de Vaan forthcoming and 2008 *passim*) for Italo-Celtic *-ei- ~ *-i-.

17 Here also OCS *kažǫ, kazati* 'zeigen', but with unexplained root variant *kʷeĝ-.

18 So García Ramón (1998: 151): "*vadh* (sekundär auch *vadhⁱ*)" (cf. aniṭ *avadhrá-* 'indestructible', vs. aor. *ávadhīt*).

ic present, but appears not to have made a root present of any kind. Other explanations of Lat. *cōnārī* are certainly available, including a R(ō)-*eh₂i̯e/o-* iterative **kōn-āi̯e-* (of the type *στρωφάω* 'twirl about', *νωμάω* 'wield', etc.),[19] or a denominative to a R(ṓ)-*eh₂* noun **kṓn-eh₂*.[20] Nor is the case for a **kṓnh₁-i̯e-* strengthened by the other forms placed here (from Celtic and Slavic), as these involve "Suffixersatz".

In a general way, of course, "Suffixersatz" is a necessary part of the theory, since in some branches – not only Germanic (**2.2**, *ad fin.*), but also Celtic, Albanian – **-éi̯e/o-* and plain **-i̯e/o-* fall together. Items like the following therefore cannot prove the existence of LIV's type "caus.-it. R(ṓ)-*i̯e/o-*":

6 caus. **i̯ṓs-i̯e-* (: **i̯es-* 'sieden, schäumen'): here only ON *æsa* 'in heftige Bewegung setzen';
7 iterat. **kṓlh₁-i̯e-* (: **kelh₁-* 'verlocken etc.'): here only ON *hœla* 'rühmen, preisen', OHG *huolen* 'betrügen' etc.;
8 caus. **(s)gʷṓsh₂-i̯e-* (: **(s)gʷesh₂-* 'erlöschen'): here only OCS *u-gasiti* 'auslöschen'.[21]
9 iterat. **kʷṓlh₁-i̯e/o-* (: **kʷelh₁-* 'eine Drehung machen'): here only Gk. *πωλέομαι* 'come and go' [Hom.], beside well-attested **kʷolh₁-éi̯e/o-*.

Most such forms, then, are indistinguishable from those with lengthened *ō*-grade and **-éi̯e/o-* suffix (like Gk. *ὠθέω*), or, in theory, lengthened and accented *ō*-grade with **-ei̯e/o-* suffix. Indeed, the *-s-* of ON *æsa* ((6) above), without Verner's Law, suggests **i̯ṓs-ei̯e-*, with root accent.

In **su̯ṓp-i̯e-* itself and the alleged R(ṓ)-*i̯e/o-* forms (1) through (9) above, we have seen all but two of the dozen acrostatic causative forms that LIV registers as "certain":

10 iterat. **pṓth₂-i̯e-*, in *πωτάομαι* 'fly' (LIV s.v. 2. **peth₂-*);
11 caus. **kʷṓi̯-i̯e-* (LIV 1. **kʷei̯-* 'wahrnehmen, bemerken') in Ru. *káju, kájat'* 'blame' (*kájat'sja* 'repent', cf. OCS *kajǫ sę, kajati sę*).

19 On this type and its background, see now Villanueva Svensson (forthcoming), with earlier references.
20 Vine 1998: 690 n. 21, similarly Tremblay 1998.
21 And again (LIV): "**-gas-je-* mit sekundärer Flexion nach dem produktiven Kausativtyp auf *-iti*".

Alternative accounts are available for both of these. A full discussion would take us too far afield, but in brief: for (10), *πωτάομαι* is equally plausible as a lengthened-grade iterative in **-eh₂i̯e/o-* (cf. (5) above on the type *στρωφάω*); and for (11), the assignment to 1. **kʷei̯-* (on morphological grounds, since 1. **kʷei̯-* makes an acrostatic present) is unconvincing, as opposed to the traditional (and semantically more straightforward) assignment to 3. **kʷei̯-* 'blame, punish etc.', which opens the door to other options, such as a lengthened-grade present of the type *πλώω* 'swim', OE *flōwan* 'flow' (cf. Jasanoff 2003: 224 n. 2). We may note, finally, that the attempt by Isebaert (1992) to add additional Latin "*sōpīre* forms" (*mōlīrī* 'strive', etc.) has been convincingly refuted by Martzloff (2006: 679ff.).[22]

The interim conclusion to be drawn from this survey is that the best example of a Klingenschmitt/LIV-style lengthened *ō*-grade iterative-causative – and possibly the only truly unambiguous-looking example – may be Lat. *sōpīre* itself. This is, for one thing, too slender a basis on which to establish an entire PIE category of the type hypothesized by Klingenschmitt and codified by LIV, even if (as seems likely) some type of special acrostatic iterative-causative may indeed have existed. At the very least, we are authorized to consider alternative accounts of Lat. *sōpīre*.

2.4 Some older ideas[23] are no longer acceptable: for example, Brugmann's conception of *sōpīre* as denominative to an *i*-stem noun **su̯ōp-i-* cannot stand, since there is no such noun, and no evidence that such an *i*-stem type ever existed.

Considerably more attractive, and nowadays rather widely accepted, is Nussbaum's conception of *sōpīre* as denominative to a root noun **su̯ṓp-s*.[24] In more detail: although Lat. *sopor* '(deep) sleep' is morphologically ambiguous,[25] Nussbaum suggests that the widespread thematic *no*-stem (with fluctuating root vocalism) **su̯op-no- ~ *su̯ep-no- ~ *sup-no-* could point to an old root noun. For the semantics: we would have an agentive root noun

22 On *mōlīrī* (as denominative), see now also Schaffner 2006: 23.

23 See the survey by Martzloff (2006: 677, with references).

24 Nussbaum 1975: 129, 1976: 250 (similarly Bammesberger 1980: 6, as a possibility); followed, for example, by Martzloff 2006: 678, de Vaan 2008: 575.

25 The main candidates are an animate *s*-stem **su̯ép-ōs* or an *r/n*-stem **su̯óp-r̥* (see de Vaan 2008: 575, s.v., for discussion).

(formally of the type *κλώψ* 'thief'), with a meaning 'sleep' – not as an activity, but as a force that induces sleep; thus the causative (or rather "pseudo-causative") meaning of the denominative would be an automatic effect of this particular agentive value. Nussbaum compares, finally, Lat. *custōdīre* 'to guard', a denominative in *-īre* to the agentive consonant stem *custōs* 'guard'.

As attractive as this seems, it is not without problems. As for the alleged root noun **su̯ŏ́p-s*: the thematic *no*-stem forms (**su̯op-no-* etc.) could equally well derive from an *r/n*-stem, as Schindler thought (1966), and as others still think (e.g. Pinault 2009: 243f.);[26] and in any case, such *no*-stems with fluctuating ablaut (also e.g. **u̯es-no-* ~ **u̯os-no-* ~ **u̯ēs-no-* to **u̯es-* 'sell', among others) constitute a general problem, with no clear basis in root nouns (as opposed to other formations, such as *n*-stems or *r/n*-stems). Nor is a nominal **su̯ōp-* unambiguously attested, in Italic or anywhere else. The proposed semantic interpretation is exceedingly clever, and seems possible; but one would like to have parallels for agentive root nouns of this type, where a notional starting point 'sleeper' amounts to 'sleep-inducer' or 'sleep-effectuator', with something like factitive value.[27] As for the parallel with Lat. *custōdīre*: this may provide only limited support, since the rare consonant-stem denominatives in *-īre* seem mainly to be analogical creations (see Leumann 1977: 556), and thus need not reflect an old derivational type as such. Finally, this account leaves ON *sœfa* out of the picture; yet, as we have said, the formal and semantic convergence between this form and Lat. *sōpīre* is striking, and potentially important for the morphological background of both forms.

2.5 In pursuing an alternative analysis of Lat. *sōpīre*, it is reasonable to operate with the same basic assumption that drove Klingenschmitt's pioneering study, namely that the verb does indeed reflect an old present formation – indeed, a causative with lengthened-grade root, which may even be a lengthened-grade causative associated with an original acrostatic root

26 In addition to Lat. *sopor,* cf. the *r*-stem presupposed by the Hittite denominative *šuppariia-*^{zi} 'sleep'.

27 On 'sleep' as an 'acting force' and 'the power which puts to sleep': see now also Pinault 2009: 245; but his discussion is based on forms belonging to the supposed *r/n*-stem, rather than an agentive root noun.

present. Since, however, at least some lengthened-grade causatives seem to have been formed beside other types of present formations (cf. the LIV tallies in **2.2.**), this last point is non-essential. More important is the idea that lengthened-grade causatives of the Klingenschmitt/LIV R(ṓ)-*i̯e/o*-type, with zero-grade suffix as compared with canonic iterative-causatives in **-ei̯e/o*-, may never have existed as such, as I argued above. Or if they did (and resulted, for example, from a very early reduction process), no clear trace of such forms actually survives, other than the *explanandum sōpīre* itself. Instead, one can posit the existence (or the early development) of acrostatic iterative-causatives with lengthened *ō*-grade root and root accent (similar to acrostatic root presents in **-ē*-), but with the ordinary iterative-causative suffix **-ei̯e/o*- (i.e. R(ṓ)-*ei̯e/o*-). Inevitably, most forms of this type would have been remodeled with the suffix accent of ordinary iterative-causatives: hence forms like Gk. *ὠθέω*, *πωλέομαι* (**2.3**(4), (9)) or ON *nœra* 'revive' (< **nōs-éi̯e/o*-[28]) with "Akzentersatz" – a less drastic and more credible adjustment than the pervasive "Suffixersatz" postulated by followers of the Klingenschmitt/LIV theory. The final assumption that must be made is that (as in the various formulations surveyed in **1.2** above) Italic may have inherited at least some traces of PIE mobile accent.

2.6 It is clear, first of all, that an original **su̯ōp-éi̯e/o*- – that is, the remodeled acrostatic type with reaccentuation (as in forms like *ὠθέω*) – would have ended up as Lat. ×*sōpeō,* ×*sōpēre,* like *monēre* 'advise' and other ordinary causatives. Yet *sōpīre* is functionally causative, has an isolated lengthened *ō*-grade, and stands a good chance of being a relic form, possibly the only relic of a lengthened-grade causative that is certifiably distinct from R(ō)-*éi̯e/o*- causatives. We may hypothesize, then, that *sōpīre* continues the more original acrostatic causative, i.e. a root-accented **su̯ṓp-ei̯e/o*- – accentually distinct from both ordinary causatives like **mon-éi̯e/o*- and reaccented acrostatic causatives of the type *ὠθέω*.[29] We can thus rephrase the problem as follows: an original **su̯ṓp-ei̯e/o*- develops not to ×*sōpēre*, but rather to *sōpīre,* exactly as if it were **su̯ṓp-ii̯e/o*-.

As an initial step, I propose (in a way that will at first seem completely *ad hoc*) the following purely phonological solution, i.e. an early Italic rule

28 Jasanoff 2003: 224, Malzahn 2007: 237.

29 ON *sœfa* might have the same background, but is simply ambiguous (**2.2**).

×́-e.i̯V > ×́-i.i̯V (in informal notation), which would state that unaccented /e/, in the syllable after an initial accented syllable, is raised to /i/ when it precedes /i̯/. Such a rule – a typologically natural assimilatory neutralization for unstressed syllables – would directly convert an original **su̯ó̯p-ei̯e/o-* to **su̯ó̯p-ii̯e/o-*.[30] Note further that this rule must be assigned to a very early stage of Proto-Italic, as it precedes the loss of intervocalic **-i̯-* (itself normally assigned to Proto-Italic: so e.g. Meiser 1998: 91). Within Latin itself, one can also point to some similar processes involving vowel assimilation before /i̯/. At an early stage, the so-called "*pius*-Gesetz"[31] fronts /ū/ to /ī/ before immediately-following syllable-initial /i̯/ (although accent is not a factor). At a later stage, an even closer match (depending on how it is conceived phonetically) might be seen in the process that accounts for pronominal forms like MIEIS 'meis', IVM 'eum', IEI 'eī', and more generally the common late Republican and Imperial variants *iī, iīs* etc. (for *eī, eīs* etc.).[32] Here, in fact, accentual conditioning may be involved, given the proclitic and enclitic behavior of possessives like *meus,*[33] and the frequent "Wackernagel" behavior of unstressed pronominal forms like *eum*.[34]

Having now proposed a phonological treatment of the sequence **-ei̯V-*, under particular accentual conditions, designed to account for Lat. *sōpīre,* it behooves us to consider other instances of inherited **-ei̯V-* in Latin (and elsewhere in Italic), so as to evaluate the behavior of our rule in those cases as well. Two such cases come to mind, namely so-called "material adjectives" in Lat. *-eus* < PIE **-ei̯o-* (Lat. *aureus* 'golden', etc.), and the nom. pl. of *i*-stems (< PIE **-ei̯-es*), also generalized in Latin as the nom. pl. of consonant stems. But it will be most convenient to return to this material after treating the second problematic morphological category of interest, namely *o*-stem denominatives in *-īre/-īrī*.

30 For recent theoretical work on vowel-quality neutralization (vowel reduction) see e.g. Crosswhite 2001, 2004, Flemming 2005, and Barnes 2006; for applications to Latin and Italic, see Nishimura 2008.

31 See recently Weiss 2009: 142, 468.

32 Cf. also O. *ioc, ionc, iusc,* Marruc. *iafc* etc. (< **ei̯ā-k(e)* etc.).

33 Especially voc. (Plt. *mī fīlī* and *gnāte mī,* etc.), cf. also monosyllabic fem. *mea* (in early drama), whether by synizesis (implying /mya/, cf. MIEIS etc.) or already with Vulg. Lat. "reduced" *mo-* (see e.g. Leumann 1977: 465).

34 See Adams 1994, 1996.

3 Italic *o*-stem denominatives

3.1 The formation of standard denominatives in Indo-European is clear and well-known: the nominal stem is suffixed with accented *-*i̯é/ó-*, as in a familiar example like Ved. *pr̥tanā-* 'battle' → *pr̥tanāyánt-* 'doing battle, fighting'.[35] There are, to be sure, various patterns of secondary development; but these tend to fall into certain well-documented subtypes, such as the generalization of resegmented sequences as productive denominative morphemes,[36] or reaccentuation (in particular based on iterative-causatives in *-*éi̯e/o-*).[37]

For denominatives made to *o*-stems specifically: these follow the usual pattern; and where the stem vowel can be determined, it shows *-*e-* preceding the suffix, and not *-*o-* (R-*e/o-* → R-*e-i̯é/ó-*), as clearly in Greek: *κόσμος* 'order' → *κοσμέω* 'arrange', etc. So also in Armenian, as in a form like *gorcem* 'work, do' (cf. *gorc* 'work'), where Arm. *-em* is in principle ambiguous between denominative *-*e-i̯é/ó-* and iterative *-*éi̯e/o-*, but in any case shows *-*e-* and not *-*o-*. Denominative *-*e-i̯e/o-* is also likely for Luvian and Lycian (though with reaccentuation based on iterative-causatives, as in Sanskrit; see Melchert 1997: 134ff. for details). It is evident, finally, that no information about the quality of the thematic vowel is recoverable from patterns in which the vowel itself is "submerged", so to speak, as a result of phonological or morphophonemic processes, as in Germanic Weak II denominatives (Go. *hails* 'healthy' → *hailjan* 'heal', etc.); cf. the parallel "submerged" *-*e-* of Germanic causatives, already mentioned above in connection with ON *sœfa* (**2.2**).

3.2 There is a range of Latin material that one might loosely refer to as "*o*-stem denominatives", including the following:

35 On PIE denominatives, see recently Alfieri 2008, Schaffner 2008.

36 E.g. Ital. *eh₂*-stem denominatives → productive denominative formant *-*āi̯e/o-*, or Gk. *id*-stem denominatives → productive *-ιζε/o-*.

37 As with Vedic *o*-stem denominatives in *-áyati* instead of *-ayáti* (Jamison 1983: 50ff., 70ff., 179f.).

1 *o*-stem denominatives formed with productive resegmented *-*āi̯e/o*- (n. 36 above), on which see *in extenso* Steinbauer 1989;[38]
2 *-*āi̯e/o*- verbs built to adjectival bases of the type *novus* 'new' → *renovāre* 'make new again' – but these factitives are only superficially similar to denominatives *stricto sensu,* and have a different morphological background (see Jasanoff 2003: 139ff.);
3 *-*ēi̯e/o*- verbs built to *o*-stem adjectives of the type *albus* 'white' → *albēre* 'be white' – but these are "denominative statives" in *-*eh₁-i̯e/o*- (built with the stative morpheme *-*eh₁*-) and thus not "standard denominatives" of the type surveyed in **3.1** above.[39]

In contrast to all of these stands the following category:

4 *o*-stem noun and adjective bases with pendant denominative in *-īre/-īrī*: e.g. *servus* 'slave' → *servīre* 'be a slave', *saevus* 'cruel' → *saevīre* 'be cruel'. This is the category of interest here, as it is acknowledged to be the repository of "true" *o*-stem denominatives, comparable to Gk. *κοσμέω* etc. (**3.1**).[40] The formation is also attested with certainty in Sabellic, and thus old within Italic: corresponding formally to Lat. *servīre* is U. **seritu**/*seritu* 'observato; servato', i.e. /serritu/ < **seru̯ītōd*.[41] As for the semantic discrepancy: as Rix has argued,[42] Lat. *servīre* reflects a semantic renewal based on the late specialized meaning ('servant, slave') of *servus,* originally 'guardian' *vel sim.*, whence U. **seritu** 'servato' (→ 'observato'). Nor is this the only instance of a Sabellic *o*-stem denominative in *-ī-*: leaving aside less clear possibilities, a highly probable case appears in U. *stahmito* 'established', denominative to (dat. or loc. sg.) *stahmei* 'establishment,

38 Note also Schaffner 2008 on the pattern "abstract/collective *-*eh₂*- → *-*eh₂-i̯é/ó*-".
39 A somewhat different conception of the background of denominative statives is found in the work of Olsen cited towards the end of **3.3**(1) below.
40 Forms in this category are listed by Mignot (1969: 59); note also the recent discussion by Martzloff (2006: 688ff.).
41 The development *-*ru̯*- > U. /-rr-/ <r> is now confirmed by Pre-Samnite *οσερρια[* (Ps 20; 6th/5th c.) ~ U. **anzeriatu**/*anseriato* (see Martzloff 2006: 633f.).
42 1994: 69ff., see also Meiser 2003: 69.

agreement', best taken as an *o*-stem.[43] The longstanding problem at issue is the descriptive fact that denominatives of this type behave as if they were based on *i*-stems: (*servus* →) *servīre,* like (*sitis* 'thirst' →) *sitīre* 'be thirsty'. This puzzling behavior has never been satisfactorily explained, as the following survey of previous approaches should make clear.

3.3 There are essentially three analytical gambits that have been proposed:

1 According to Nussbaum (1975: 149f.), the *o*-stem denominatives in *-īre/-īrī* contain an inherited long **-ī-,* which is historically identical to the *o*-stem gen. sg. formant *-ī* (< **-iH*) that also appears in derivatives like (*servo-* ~) *servī-lis* (an archaic *o*-stem pattern: also *vir* 'man' ~ adj. *virīlis, erus* 'master' ~ adj. *erīlis,* and several others), beside *servī-re*. The denominative **-ī-i̯e/o-* formation – originally proper to nouns – would then have been extended, rather early, to thematic adjective bases, whence forms of the type (*saevus* →) *saevīre,* (*blandus* 'flattering' →) *blandīrī* 'caress, flatter', etc.

There is some semantic attractiveness to this theory, insofar as most of the forms in question have "essive" meaning: i.e., *servīre ~ servus esse, servīlis esse* (like *saevīre = saevus esse*, *blandīrī = blandus esse*). Yet there are disquieting features as well. In formal terms, there is no trace of such a pattern of verbal derivation elsewhere, with the marginal exception of the Vedic type *putrá-* 'son' ~ RV *putrīyánt-* 'desiring offspring' (also AV *putriyánt-*), which has long been compared to a small subset of Latin denominatives with somewhat similar semantics, like *equīre* 'be in heat' (as if 'desire the stallion'), *catulīre* 'id.' (cf. *equus* 'horse', *catulus* 'puppy, whelp'). Nevertheless, as I discuss elsewhere (2009), the Vedic type (which, among other earmarks of late formations, has no parallel in Avestan, and may

43 Hamp 1985, followed by Untermann 2000: 696; so also García Castillero 2000: 187 n. 177 ("probable formación nominal en *-mo-s*"), who observes further that the rationale for *ī*-conjugation is unclear. His speculation about influence from the semantically similar U. **statita** is unconvincing, and needlessly separates (*stahmo-* ~) *stahmi-* from (Lat. *servo-* ~) U. **seri-**/Lat. *servī-* and the entire complex of *o*-stem denominatives with *ī*-conjugation.

even go back to *-*i*-*i̯e/o*-) is likely to reflect an inner-Indic development. Nor is it apparent why this "genitival" pattern in *-*ī*-*i̯e/o*- would have replaced inherited *-*e*-*i̯e/o*-, while the inherited *ā*-stem, *i*-stem, and *u*-stem patterns remain intact (i.e., *-*ā*- → *-*ā*-*i̯e/o*-, *-*i*- → *-*i*-*i̯e/o*-, *-*u*- → *-*u*-*i̯e/o*-), with no additional layer of derivation. Moreover, as Nussbaum admits, there is at least one old-looking item that is not essive semantically, namely *artīre* 'tighten, fit (s.th.) tightly (onto s.th.)' [Cato+], cf. *artus* 'tight'; and it may be possible to add other forms of this type. Note, for example, Eichner's analysis of Lat. *(red)hostīre* 'requite, compensate' [Plt., Naev.+] beside *hostus* 'yield from olive-pressing' [Cato, Varro] < *'substitution, compensation' (2002, cf. Martzloff 2006: 690). Further, although the problem cannot be examined here in full, another such non-essive instance may be Lat. *polīre* 'polish' [XII Tab.+], if Vendryes (1929) was correct in taking this as derivationally related to *interpolis* 'refurbished' [Plt.+] and if the thematic variant *interpolus* [glosses, Pliny] is actually the older form, as de Vaan argues (2008: 477).[44] Finally on this point, the essive (or "stative") meaning of *servīre* itself may even be secondary (cf. **3.2**(4) above on Rix's account of the semantics of U. **seritu**/*seritu* vis-à-vis Lat. *servīre*); and in any case, PIE denominative formations regularly incorporated both essive and factitive options (cf. Tucker 1988: 98, Schaffner 2008).

An account that is formally similar to that of Nussbaum (though with a different morphological conception) has been proposed by Olsen (1996, 2003: 235ff.). She begins with *i*-stem variants of *o*-stems, which are then suffixed with a zero-grade version of the stative suffix, thereby producing a sequence (*-*i*-*h*$_1$-*i̯e/o*-) that would be appropriate for the stative or "essive" meaning of forms like *servīre* and *saevīre*. Even if one accepts the existence of zero-grade *-*h*$_1$-,[45] this approach is subject to most of the objections just mentioned, together with others concerning the presumed *i*-stem basis, to which I return in **3.3**(3) below.

2 Another solution was proposed by Rix (1994: 71), according to whom the starting point was precisely the form *ser*u̯*-i̯e/o*-, based on

44 See de Vaan for additional details, including other possible accounts of Lat. *polīre*.

45 See, however, Jasanoff 2002/3 (*passim*), 2003: 155ff. (cf. n. 15 above).

an old *u*-stem **ser-u-*.[46] This would yield **seru̯-iu̯e/o-*, by an "Italic version of Sievers' Law"; and on this basis a generalized *o*-stem denominative suffix "**-iu̯e/o-*" would be abstracted, through a secondary association of **seru̯-iu̯e/o-* with the *o*-stem **seru̯-o-*.[47] It is, however, distinctly unattractive to explain the entire formation as having arisen from a generalization based on a single form.[48] Nor, as Martzloff observes (2006: 635), would one expect the old denominative of a *u*-stem **ser-u-* to syllabify as "**seru̯-i̯e/o-*", as Rix claimed. Rix's further suggestion that the association **seru̯-o-* : **seru̯-i̯e/o-* might itself reflect something old, with reference to "truncated" *o*-stem denominatives of the type (*ἄγγελος* 'messenger' →) *ἀγγέλλω* 'announce', raises more problems than it solves. There is so far no other evidence for such formations in Italic, and so the comparison may simply be specious (see Martzloff 2006: 689); and in any case, the PIE status of such "truncated" *o*-stem denominatives (though attested with some frequency in Greek) remains unclear.[49]

3 If one is going to suggest a basis in a thematic derivational process: rather than truncation, one might more appropriately operate with an alternation between the thematic vowel and **-i-* – and this, indeed, constitutes a popular recent approach to the problem. Note, for example, Schrijver (2003: 60), citing Lat. *forctus* ~ *for(c)tis* 'strong' and *sacro-* ~ *sācri-* 'sacred': "the original derivational basis of this type [scil. *o*-stem denominatives in *-īre*; BV] may be an *i*-stem, too. As is generally agreed, … the denominatives belonging to the fourth conjugation may be reconstructed as thematic *i*-derivatives of *i*-stem nouns and adjectives". Yet one of the weaknesses of this theory is immediately apparent from Schrijver's own examples: there

46 Somewhat similarly Rasmussen (1989: 98), who reconstructs the root itself as **serh₂u-* (with "aniṭ" variants generalized from forms that would have undergone laryngeal loss, such as **sór[h₂]u̯-o-*).

47 This conception is followed by Meiser (1998: 196; 2003: 67, 69, 135).

48 See already on this point Nussbaum 1975: 149.

49 For recent discussion, with particular attention to the Avestan material, see Tucker 2004 (differently on some forms Tremblay 2008: 24f.). Oettinger (2004: 381) argues for an Anatolian example.

is no Lat. ×*for(c)tīre* ('be strong'? 'strengthen'?) and no ×*sācrīre* ('be sacred'? 'sanctify'?).[50]

A more precise formulation of just such an approach has recently been offered by Martzloff (2006: 688ff.), explicitly based on the theory of thematic morphological derivation articulated by Schindler (1976: 351, 1980) and developed since by many others.[51] Martzloff's version operates with the following three assumptions. First, it is taken as given (following Schindler) that *o*-stem adjectives can regularly serve as a derivational basis for *i*-stem abstracts (with replacement of *-*o*- by *-*i*-). It is these *i*-stem abstracts, in turn, that serve as the basis for denominative verbs in *-*i*-i̯*e/o*- (ultimately based on thematic adjectives). Finally, this derivational chain ("adj. *-*o*- → abstr. *-*i*- → denom. *-*i*-i̯*e/o*-") can eventually be reinterpreted, by suppression of the intermediate stage, as a simpler process, whereby *o*-stem adjectives (and, secondarily, *o*-stem nouns) can directly develop denominatives in *-*i*-i̯*e/o*-, without the actual existence of intermediate *i*-stem formations.[52]

This theory is not credible, for a number of reasons. To begin with the data: Latin does indeed have several *i*-stem abstracts of the requisite type – i.e., forms ultimately interpretable as pendants to original thematic stems – and these include both nouns and adjectives. For the latter, we have already seen Lat. *fortis* (and, insofar as it may be relevant, *sācri*-), and I have discussed several other such *i*-stem cases elsewhere (2006b), including *cīvis* 'citizen' (originally *'[societal] closeness'), *collis* 'hill' (< *'elevation'), and others. There are, as noted above, no corresponding denominatives ×*fortīre* and ×*sācrīre*; and there is likewise no ×*cīvīre* ('be close'?), no ×*collīre* ('be high'? 'raise'?), nor any derived verbs in *-īre* related to any member of the original set of isolated *i*-stems (beside *o*-stems) discussed by Schindler (no ×*imbrīre* beside *imbri*- 'rain', no ×*torrīre* beside *torris* 'firebrand', etc.),

50 Lat. *sacro*- ~ *sācri*-, moreover, is anything but a simple *-*o*- ~ *-*i*- alternation, however the lengthened grade of *sācri*- (apparently [Weiss 2009: 341] a kind of vṛddhi formation, vis-à-vis *sacro*-) is to be explained.

51 For some theoretical discussion (from a range of perspectives), see recently Balles 2009: 9ff., Olsen 2009: 194f., Rasmussen 2009: 261ff., Rau 2009: 130f.

52 The *i*-stems envisioned in Olsen's theory (**3.3**(1) above) have a different etiology (see Olsen & Rasmussen 1999), but can be considered in the same terms.

apart from trivial or apparent exceptions.[53] Nor, conversely, is there a single *i*-stem form (whether an original abstract or an adjectivized version) that is found beside any attested *o*-stem denominative verb: i.e., for nouns, there is no *i*-stem ×*serv-i-* (: *servus*) corresponding to *servīre,* no ×*proci-* (: *procus* 'suitor' [Plt.+]) corresponding to *procīre** 'be a suitor, woo' (supine *procītum* [Liv. Andr.]), and for adjectives, no ×*saevi-* (: *saevus*) corresponding to *saevīre,* no ×*bland-i-* (: *blandus*) corresponding to *blandīrī,* etc.[54] Nor, despite Martzloff, is there any indication that the adjective type is older than the noun type: in addition to *servīre* and *procīre** (both evidently old), note also *fastīdīre* 'feel disgust at, scorn' [Plt.+] (: *fastīdium* 'disgust' [Plt.+]). Indeed, it is the better-represented adjective type that evidently became semi-productive, whereas the clearly residual *servīre* type has the appearance of an archaism. To put the entire matter in perspective: not only is the crucial "intermediate *i*-stem stage" bypassed, but it is in fact nowhere to be found (including, it should be noted, for the supposedly more basic adjective forms): we are asked to believe, in other words, that the entire substitutive morphological process occurred prehistorically, *without a single surviving model*. And at the same time, there is not a particle of evidence, in Italic or elsewhere, for a parallel denominative process; nor is any substantive reason provided as to why the expected formation of thematic denominatives (*-*e/o*- → *-*e-i̯e/o*-, like *κοσμέω* etc.) should have been replaced by the more complex substitutive process envisaged (*-*o*- → *-*i*- → *-*i-i̯e/o*-), when other vowel-final stems in Italic regularly follow the standard IE denominative pattern, as with *ā*-stems (→ *-*ā-i̯e/o*-), *i*-stems (→ *-*i-i̯e/o*-), and *u*-stems (→ *-*u-i̯e/o*-). At best, one might venture the claim that "homonymie fâcheuse" could have blocked the simple pattern *-*e/o*- → *-*e-i̯e/o*-, as this would have resulted in Italic "second-conjugation" *ē*-verbs. But this fails to convince: why should thematic denomina-

53 Thus *febrīre* 'be sick with a fever' [Celsus+] (cf. *febris* 'fever') is a patently late creation based directly on the Latin *i*-stem, which has no *o*-stem pendant in Latin (cf. *τέφρα* 'ashes'); and the alleged *rāvīre* 'talk oneself hoarse' at Plt. *Poen.* 778 (1 sg. *rāviō,* cf. *ravus* 'hoarse' and *ravis* 'hoarseness') may be corrupt or may be interpretable as *rāviāre* (see de Vaan 2008: 515, with references), and in any case shows a problematic lengthened grade that renders the form unusable.

54 It is highly unlikely, for semantic reasons, that *artīre* 'fasten tight' (**3.3**(1)) is denominative to *ars* (gen. *artis*) 'skill' rather than to *artus* 'tight'; cf. on this point de Vaan 2008: 56 s.v. *artus.*

tives have constituted a critical mass for avoidance of the second conjugation, when so many other formations fell together as *ē*-verbs in just the same way (i.e., stative **-eh₁-i̯e/o-*, iterative-causative **-éi̯e/o-*, athematic **Ceh₁-(i̯e/o-)*)?

There is, then, room for further consideration of the problem of Italic *o*-stem denominatives.

3.4 Clearly the simplest and most appealing approach would be one according to which the residual *o*-stem denominative type *servīre, saevīre,* etc. could be explained directly in terms of the predicted PIE pattern **-e-i̯é/ó-* (comparable to forms like Gk. *κοσμέω*). This, I suggest, can be accomplished quite simply by a minimal reformulation – indeed, a generalization – of the phonological rule posited above (**2.6**) to account for Lat. *sōpīre*: if one generalizes the raising of /ĕ/ to /i/ before /y/ so as to apply to any unaccented /ĕ/, this will explain not only the post-tonic raising in **su̯ŏp-ei̯e/o-* (> **su̯ŏp-ii̯e/o-*), but also pre-tonic raising in thematic denominatives: (Ital.) **seru̯-e-i̯é/ó-* > **seru̯-ii̯e/o-* (> *servīre*), **prok-e-i̯é/ó-* > **prok-ii̯e/o-* (> *procīre**), etc.[55] As a consequence, the Italic residual *o*-stem denominatives of the fourth-conjugation *servīre* type would directly recover the predicted PIE denominative pattern (exactly as in denominatives to other vowel-final stems: *ā*-stems → **-ā-i̯e/o-*, etc.; **3.3**), without a need for the suspiciously elaborate (and in any case entirely unsupported) derivational mechanism involving "intermediate *i*-stems". We would then, in ad-

55 Although the matter cannot be pursued in detail here, note that this analysis accounts likewise for *o*-stem denominatives based on formations with thematic suffix (rather than plain thematic vowel). Thus, if the obscure Latin formations in *-urīre* (*parturīre* 'be in labor', etc.) and *-ūrīre* (*ligūrīre* 'lick, eat daintily', etc.) are ultimately based on *ro*-stems (see Martzloff 2006: 683f., and further Fortson 2007 on the *ro*-stem basis of Lat. *-tūrus/-sūrus-*), denominative **-re-i̯é/ó-* > **-ri-i̯e/o-* would be regular, according to the accent-based formulation just suggested. So also perhaps for *audīre* 'hear', if this is denominative to an adjective (pre-Lat.) **au̯izdo-* 'making/made clear' (< virtual **h₂eu̯is-dʰh₁-o-*, cf. **m̥gh₂-dʰh₁-ó-* [> *ἀγαθός* 'good'] and other such compounds; Vine 2006c: 504f., 511), whence **au̯izd-e-i̯é/ó-* > **au̯izdii̯e/o-*. (Somewhat differently de Vaan 2008: 61 s.v., as if directly from **h₂eu̯is-* + "**dʰh₁-i̯e/o-*" 'to render'. On this general conception of *audīre,* see Viredaz 2003, with references.)

dition, have further evidence for the maintenance, in early Italic, of at least some mobile accent contrasts with PIE pedigrees.

3.5 We may now return (cf. **2.6**, *ad fin.*) to the two additional categories that may show the sequence /eiV/, so as to evaluate their interaction with the proposed accent-based formulation.

1 As already noted, the "material adjectives" in *-eus* (Lat. *aureus* 'golden', etc.; Weiss 2009: 273) are built with a suffix **-eiˌo-*.[56] According to the accent-based raising rule as formulated above, this should mean that the version of the suffix inherited by Italic had the accentual profile **-éiˌo-*, since unaccented **-eiˌo-* or oxytone **-eiˌó-* should both have produced forms like Lat. *×aurius, ×argentius* 'of silver', and the like. In fact, although the PIE situation is not entirely clear in some respects, there is reason to believe that the suffix can indeed be reconstructed in this way (so explicitly Meier-Brügger 2003: 284). Despite some variation,[57] this is certainly the pattern displayed by the core Vedic material, as in forms like (*híraṇya-*'gold' →) RV *hiraṇyáya-* 'golden', (*ávi-* 'sheep' →) RV *avyáya-* 'of, from sheep', (obl. *dru-* 'wood' →) AV *druváya-* 'wooden (box)', and others.[58] In Greek,[59] the Homeric forms are almost invariably proparoxytone (*ἀργύρεος* 'of silver', *σιδήρεος* 'of iron', *χάλκεος* 'of bronze', etc., including metrically-lengthened *-ειο-*: *δούλειος* 'of a slave', etc.), apart from *θυρεός* (scil. *λίθος*) 'door-stone', which nevertheless stands somewhat apart, as a substantivization (so Schwyzer 1939: 468). But

56 In truth, as recently shown by Nussbaum (2009), the functional profile of this formant is somewhat broader; but that is not essential here, and the term "material adjectives" can be used as a convenient label.

57 See Wackernagel-Debrunner 1954: 213f. for details.

58 Different patterns shown by some forms may be related to substantivizations (*gavayá-* 'special kind of bull' [4.21.8c], cf. *gavyáya-* 'of, from cows' for **gaváya-*; *mŕ̥gaya-* lit. 'The Beast' [name of a demon]), or may have other explanations: e.g. *pávyaya-* 'Somaseihe' (9.86.34b), an "Augenblicksbildung" based on *ávyaya-* (immediately preceding, in the same pāda), this latter with analogical accent after *ávya-* [RV 26×], beside *avyáya-* [RV 27×]; see Wackernagel-Debrunner 1954: 214, following Oldenberg, and more recently Schmitt 1997: 286, 289f.

59 See Schwyzer 1939: 467f., Chantraine 1933: 50f, Risch 1974: 131ff.

the contracted forms in Attic are perispomenon (*ἀργυροῦς* etc.), as if from uncontracted paroxytone **ἀργυρέος* etc. Thus, as Probert has suggested (2003: 64), the Attic accentuation (which matches Ved. *-áya-*) may well reflect the original pattern, with the Homeric recessive accent an innovation, however that is to be explained. If so, Lat. *-eus* (< **-éi̯o-*), far from contradicting the proposed accent-based rule, could provide a *tertium comparationis* for the accentuation of this formant.[60]

2 The *i*-stem nom. pl. sequence **-ei̯-es* provides another **-ei̯V-* context; and the fact that this normally develops in Italic to /-ēs/ (as in Lat. *i*-stems and consonant stems and Sabellic *i*-stems) should point to accented **-éi̯-es* (like **-éi̯o-* in (1) above), according to the theory presented here. This poses no real problem: *i*-stems were mainly proterokinetic, and Italic seems also to have inherited a number of hysterokinetic *i*-stems (i.e. with Lat. nom. sg. *-ēs* < **-ḗi+s*: *caedēs* 'slaughter' etc.; Klingenschmitt 1992: 115ff., Weiss 2009: 242) – in both instances, full grade of the suffix is normal, and accented **-éi̯-es* (whether original or extended from oblique **-éi̯-*) could easily have been generalized.

Yet the Italic picture is more complex, and in an interesting way. As is well known, there is an alternate nom. pl. in *-īs* (used for both original *i*-stems and consonant stems), which is much better attested than is ordinarily assumed.[61] The handbooks regard this form as having been generated secondarily, as an extension of *i*-stem acc. pl. *-īs* on the pattern of consonant-

60 PIE paroxytone **-éi̯o-* might derive further support from Benveniste's etymology of the suffix (1935: 76f.), which assumes thematization of *i*-stem oblique **-éi̯-*; but his conception is based on forms like Ved. *hr̥dayam* 'heart', Gk. *ὀστέον* 'bone', Lat. *hordeum* 'barley', which may have a different background, unrelated to PIE **-éi̯o-*. The problem is a complex one, and must be left aside here. (Some recent analysis was presented in Nussbaum 2009.)

61 See on this point Sihler 1995: 317, and especially Gerschner 2002: 128, with reference to Hodgman (1902: 298), who called attention to approximately 100 instances in the manuscripts of Plautus ("too numerous to be looked on as accidents or blunders"), in addition to the handbook notices of this phenomenon (see next footnote) that cite several examples from grammarians and the attestations from early inscriptions (these normally spelled -EIS, late 2nd c. BC+).

stem nom. pl. *-ēs* ~ acc. pl. *-ēs*.[62] This account has been questioned, in my view rightly, by Nyman (1990), in a detailed study of the problem. As he points out, documenting an observation already found in Neue-Wagener, the frequent appearance of *-īs* in consonant stems makes it "highly doubtful whether the Nom. Pl. nouns in *-īs* can be looked upon as conditioned by the Acc. Pl. *-īs*" (1990: 221). Moreover, he makes a reasonable case (1990: 222f., with earlier references) that a nom. pl. in *-īs* is marginally attested in Sabellic, as possibly in O. **fertalís** ([Rix 2002] Cp 24.5 = Ve 81.5), **uhftis** (Cm 14C.11 = Ve 5c.11), Pael. *pacris* (Pg 9.6 = Ve 213.6), Marruc. *pacris* (MV 1.1 = Ve 218.1).[63] This helps support the antiquity of the ending, for which Nyman provides an explanation in terms of PIE morphology, as derived from an *i*-stem paradigm with "open inflection" and nom. pl. **-i(i̯)-es,* i.e. with zero grade of the suffix. Weiss (2009: 245 n. 21) offers two objections to this theory, of which only one may be cogent. While it is true that "there is no clear evidence that **-ii̯es* would syncopate in a closed syllable", neither is there any evidence against Nyman's proposal that the "*fīlī* rule"[64] should be generalized to include both **-i(i̯)e* and **-i(i̯)es* (1990: 226). Far more serious is Weiss's counterargument that "no other IE language has a zero-grade suffix in the nominative plural of *i*-stems". Even if this is not strictly accurate descriptively,[65] the objection stands, in terms of historical morphology. We can, however, address this point and at the same time salvage the essence of Nyman's otherwise attractive theory by proposing that the Italic nom. pl. in *-īs* goes back to an acrostatic nom. pl. in **-ei̯-es,* with root accent (thus nom. pl. **h₂óu̯-ei̯-es* 'sheep', etc.). Application of the raising

62 Thus e.g. Ernout 1953: 54, Leumann 1977: 440 (*-īs/-īs* as "hypercorrect"), Meiser 1998: 138, Weiss 2009: 245; somewhat differently Sihler 1995: 317 ("syncopated form of **-ei̯es*", given inscriptional -EIS, but with the *survival* attributed to acc. pl. *-īs* beside consonant-stem *-ēs/-ēs*).

63 As Weiss points out (2009: 245 n. 21) and as Nyman himself admits, O. **aídilis** (Po 1.2, 11 = Ve 8.2, 11), as a loanword from Latin, is "not an independent witness".

64 Syncope of **Cii̯V* to /Cii̯/ (> /Cī/), word-finally as in voc. **fīlie* 'O son' > *fīlī* (Weiss 2009: 122).

65 Nyman (1990: 224) cites forms like Hom. *ŏιες* 'sheep'; but this is more likely an innovation for **ŏ(ϝ)εες*, see e.g. Schwyzer 1939: 573, Chantraine 1961: 86. Similarly, as pointed out to me by Thomas Olander, the apparent **-ii̯es* of Slavic *i*-stem masc. nom. pl. *-ĭje* probably reflects a secondary development.

rule to unaccented *-*e̯i̯-es* will yield *-*i̯i̯-es,* whence nom. pl. *-īs* (assuming syncope in the final syllable). In this way, the alternation between the Italic nom. pl. formants *-īs* (marginal) beside standard *-ēs* is neatly explained on the basis of early Italic survival of acrostatic *i*-stem inflection (marginally) beside more widespread proterokinetic and hysterokinetic patterns.

4 Concluding remarks

I have suggested that two morphologically unrelated instances of anomalous "*ī*-conjugation" in Italic (Lat. *sōpīre* and the category of Italic *o*-stem denominatives) can both be accounted for by a typologically natural reduction rule, which raises /e/ to /i/ in unaccented position before /i̯/. The rule is not only sensitive to accent, but correlates more specifically with accent patterns that are predictable on the basis of probable (in the case of Lat. *sōpīre*) or well-established (for thematic denominatives) configurations of PIE accent. This conception, if correct, would provide further evidence, to be considered together with other such hypotheses (**1.2**), for the maintenance in early Italic of at least some form of PIE mobile accent, preceding the period of initial stress.[66] More specifically, early Italic would be seen to preserve a series of accentual contrasts in both verbal and nominal morphology:

1 root-accented acrostatic causatives in *-*e̯i̯e/o-* (with Lat. *sōpīre* as an isolated instance of this type) vs. standard iterative-causatives in *-*éi̯e/o-* and thematic denominatives in *-*e-i̯é/ó-*;
2 material adjectives in *-*éi̯o-* and *i*-stem nom. pl. *-*éi̯-es* with generalized (proterokinetic/hysterokinetic) suffix accent, vs. marginally-preserved *i*-stem (acrostatic) nom. pl. *-*e̯i̯-es,* with root accent and unaccented suffix.

In a broader perspective, one may thus conceive of Proto-Italic as a stage of Indo-European similar to Proto-Germanic, in terms of the behavior of in-

66 I intend to treat elsewhere yet another proposal regarding early Italic mobile accent, based on data pointing to a prehistoric process of "initial-syllable syncope", dependent on non-initial accents that correlate with PIE patterns. A preliminary version of that research was presented as Vine 2008.

herited accent and the development of accentual innovations: in both branches, the old mobile accent appears nowhere on the surface, having been replaced at a very early period by an initial-stress accent; yet traces of its existence at an earlier stage are visible on the basis of segmental phonological alternations that are sensitive to accent, as with Verner's Law (in Germanic), and – granting at least some of the material surveyed and proposed above – Italic phenomena such as the double treatment of C̥RHC sequences, Thurneysen-Havet's Law, and the /e/ to /i/ raising that may account for both Lat. *sōpīre* and Italic *o*-stem denominatives.[67]

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